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Wanrong (Zoey) Zheng

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Education

University of Southern California, Ph.D. student in Computer Science, Advisor: Prof. Laurent Itti — *May. 2024 — Present*
University of Southern California, M.S. in Computer Science, Advisor: Prof. Ram Nevatia — *Aug. 2021 — May. 2023*

Research Interests

LLM Spatial Intelligence [1, 2]: Developing language-guided agents that understand and navigate complex visual environments.
Multi-Modal Perception [3, 4, 5, 6, 7]: Exploring the synergistic effects of diverse data for enhanced machine understanding.

Publications

1. Zixiang Xu, Zirui Song, **Wanrong Zheng**, et al. **Grid-Bench: Can LLMs Predict How Discrete Worlds Change?** *In submission.* [paper][code][data].
2. **Wanrong Zheng**, Yunhao Ge, and Laurent Itti. **Three-Step Nav: A Hierarchical Global–Local Planner for Zero-Shot Vision-and-Language Navigation**. *Annual Conference on Artificial Intelligence and Statistics (AISTATS'26).* [paper][code].
3. **Wanrong Zheng***, Haidong Zhu*, Zhaoheng Zheng, and Ram Nevatia. **GaitSTR: Gait Recognition with Sequential Two-stream Refinement**. *IEEE Transactions on Biometrics, Behavior, and Identity Science (TBIOM'24).* [paper][code].
4. Haidong Zhu, **Wanrong Zheng**, Zhaoheng Zheng, and Ram Nevatia. **ShARc: Shape and Appearance Recognition for Person Identification In-the-wild**. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV'24).* [paper].
5. Haidong Zhu*, **Wanrong Zheng***, Zhaoheng Zheng, and Ram Nevatia. **GaitRef: Gait Recognition with Refined Sequential Skeletons**. *IEEE International Joint Conference on Biometrics (IJCB'23),* **(Oral)**. [paper][code].
6. Haidong Zhu, Zhaoheng Zheng, **Wanrong Zheng**, and Ram Nevatia. **CAT-NeRF: Constancy-Aware Tx^{2}Former for Dynamic Body Modeling**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW'23).* [paper][code].
7. Xiaoke Jiang, Yu Qiao, Junjie Yan, Qichen Li, **Wanrong Zheng**, and Dapeng Chen. **SSN3D: Self-Separated Network to Align Parts for 3D Convolution in Video Person Re-Identification**. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI'21).* [paper].

Employment

University of Southern California — **Los Angeles, CA**
Research Assistant, Advisor: Prof. Laurent Itti — *May. 2024 — Present*
Research Assistant, Advisor: Prof. Ram Nevatia — *Jan. 2022 — May. 2024*

SenseTime Technology Co., Ltd. — **Shenzhen, China**
Research Engineer, Advisor: Dr. Xiaoke Jiang & Dr. Ding Liang — *Sep. 2019 — Aug. 2021*

Awards & Honors

1st on MS1M dataset in Masked Face Recognition Challenge (ICCV 2021) out of 136 teams — *Oct. 2021*
2nd on Glint360k dataset in Masked Face Recognition Challenge (ICCV 2021) out of 86 teams — *Oct. 2021*

Service

Conference Reviewer: CVPR 2026, ECCV 2026, NeurIPS 2026, AISTATS 2026-2027
Journal Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE Transactions on Multimedia, IEEE Transactions on Cognitive and Developmental Systems, International Journal of Computer Vision

Research Experience

iLab, University of Southern California — **Los Angeles, CA**
Research Assistant, Advisor: Prof. Laurent Itti — *May. 2024 — Present*

- **Three-Step Nav: A Hierarchical Global–Local Planner for Zero-Shot Vision-and-Language Navigation**
  - Proposed a hierarchical global-local framework that alternates between *looking forward* to sketch global plans, *looking now* for fine-grained local grounding, and *looking backward* to audit execution history.

- Designed an adaptive judge module equipped with meta-skills to detect navigation drift and trigger self-correction without task-specific fine-tuning.
- Achieved state-of-the-art zero-shot performance on R2R-CE and RxR-CE datasets, reducing navigation error by 15% and improving SPL by 12% on R2R-CE validation splits.
- One first-author paper accepted at AISTATS 2026 [2].

IRIS Computer Vision Lab, University of Southern California **Los Angeles, CA**
Research Assistant, Advisor: Prof. Ram Nevatia *Jan. 2022 — May. 2024*

- **ShARc: Shape and Appearance Recognition for Person Identification In-the-wild**
  - Developed a multimodal framework that integrates the pose and shape encoder with the aggregate appearance encoder for robust person identification in uncontrolled environments.
  - Achieved state-of-the-art results on public cloth-changing person re-identification datasets such as CCVID, MEVID, and BRIAR.
  - Published one co-author paper on WACV 2024 [4].
- **GaitRef: Gait Recognition with Refined Sequential Skeletons Knowledge Exchange**
  - Combined the silhouettes and skeletons information and refined the framewise joint predictions for gait recognition.
  - On Gait3D, the proposed method outperformed the baseline by 6.1% on Rank-1 and 5.4% on Rank-5.
  - Published one primary-author paper on IJCB 2023 (oral) [5].

Identity Verification, SenseTime **Shenzhen, China**
Research Engineer, Advisor: Dr. Ding Liang & Dr. Yichao Wu *Jan. 2021 — Aug. 2021*

- **Large-scale Phone Unlock Facial Verification**
  - Updated face unlock models for Chinese mobile phone manufacturers such as Huawei, Oppo, and Vivo.
  - Prepared three different size levels of models for various products' performance needs and used different training strategies.
  - Achieved 1e-6FAR@recall 90% for different races, including Caucasian, African, Asian, Indian, and Latino.

Smart City Group, SenseTime **Shenzhen, China**
Research Engineer, Advisor: Dr. Xiaoke Jiang & Dr. Junjie Yan *Dec. 2019 — Dec. 2020*

- **Self-Separated Network to Align Parts for 3D Convolution in Video Person Re-Identification**
  - Trained the Self-Separated Network in supervised / semi-supervised / unsupervised ways, which proved the efficiency of the semi-supervised alignment strategies, which used the labels with the selected position.
  - Received a 15.5% Rank-1 improvement on iLIDS compared to the fully supervised way.
  - Published one co-author paper on AAAI 2021 [7].

Patents

1. **Wanrong Zheng**, and Xiaoke Jiang. **A Identification method Based on History Passenger Flow Big Data.** *CN113269129A* (2021)
2. Sun Zhe, **Wanrong Zheng**, Xiaoke Jiang, Xinghua Yao, and Cong Ji. **Passenger Illegal Handing Bags Across Railing Detection in Real Railway Scene.** *CN112818844A* (2021)
3. **Wanrong Zheng**, Xiaoke Jiang, Jikui Bao, Qichen Li, and Cong Ji. **A Railway Face Recognition Solution Based on History Passengers' Riding Pattern.** *CN112232424A* (2020)